

Electromagnetic Design and Velocity Regulation of BLDC Motors Using Simulation and Control Software

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ABSTRACT

A BLDC motor finds applications in automotive systems, aircraft, and robotics owing to its good control characteristics. However, the motor model designed in the simulation environment for HIL testing often requires modification because the mathematical model does not fully capture the mechanical and thermal dynamics of the system. The FEA model of the motor provides dynamic electromagnetic and thermal characteristics. This study focuses on the closed loop velocity regulation of a BLDC motor simulated for an electromechanical actuator (EMA) application to overcome modeling difficulties. The BLDC motor model developed using JMAG-RT exhibits the required thermal characteristics and is exported to the Simulink environment for speed control. A PI controller was implemented in the control loop to achieve the desired speed while maintaining a simple system. While PID control is a well established technique, the contribution of this work does not lie in the development of a new controller, but in its effective application within an FEA based BLDC motor framework developed using JMAG-RT. The study presents a cascaded dual-loop control structure integrated with a high fidelity motor model that includes electromagnetic and thermal characteristics. Furthermore, the work focuses on system-level validation under varying operating conditions, demonstrating improved dynamic performance and practical applicability for electromechanical actuator systems, thereby contributing beyond conventional PID based implementations. The proposed system exhibits good dynamic characteristics of the motor, as the model was created in JMAG with consideration of both the motor's electrical and mechanical constraints.

Keywords: Brushless DC motor; Feedback control; JMAG-RT; PID controller; Back EMF; Speed control

NOMENCLATURE

EMA	: Electromechanical Actuator
FEA	: Finite Element Analysis
JMAG-RT	: Electromagnetic Field Analysis Software
PI	: Proportional-Integral-Derivative
IoT	: Internet of Things
PWM	: Pulse Width Modulation
NLG	: Nose Landing Gear
BLDC	: Brushless Direct Current Motor
PID	: Proportional Integral Derivative
MSC Adams	: Mechanical System Simulation software by MSC Software
LVDT	: Linear Variable Differential Transformer
PSO	: Particle swarm optimization
ANN	: Artificial Neural Networks.

1. INTRODUCTION

In recent years, the demand for motors with high power to weight ratios for applications involving electromechanical actuators (EMA) has increased. The Brushless DC Motor (BLDC) is highly energy efficient, and its size to efficiency

ratio is also reasonable for more reliable applications, such as steering wheel control in aircraft. Control of the BLDC motor with precision requires a carefully designed set of motor parameters and controllers. The motor model designed in the simulation environment using the first principles technique lacks mechanical and thermal constraints. Thus, the output characteristics of the motor model vary when the design of the same motor is manufactured or during hardware in the Loop testing. This is due to the nonlinearity in the motor, which cannot be simplified by designing a motor with only fundamental Eqn., as electrical and mechanical constraints are not inherited. This constraint must be included for an output closer to the reference when the motor is manufactured.

The research significance in obtaining an efficient motor design has increased recently. Because prototyping a motor with reasonable cost and precise control is both time consuming and expensive, simulation provides approximate results before manufacturing a prototype, and modeling and simulation for speed control application of BLDC motor¹ tends to provide similar results. The motor model in the Finite Element Analysis (FEA) provides electromagnetic characteristics, and flux linkage² is effective for motor design. Complex control techniques are employed for precision, and soft computing techniques, including Fuzzy, Particle Swarm

Optimization (PSO), and Artificial Neural Networks (ANN), have been proposed to control the speed of the BLDC motor³, thereby increasing accuracy. The motor model used here is the IPM of different rotor and stator configurations, which can be optimized to the required slot/pole configuration in the JMAG environment⁴. The present work extends beyond existing studies by integrating the high-fidelity FEA model with a cascaded dual-loop control architecture specifically designed for electromechanical actuator (EMA) applications. Unlike prior work, the proposed approach incorporates coupled electromagnetic thermal characteristics within the control framework and evaluates system performance under varying load conditions. This system level integration, along with real time compatible model export to MATLAB/Simulink and detailed dynamic analysis constitutes the primary novelty and contribution of this study.

The 12S-14P BLDC motor model, optimized for drone applications, is referred to as such. After modeling the motor in FEA, JMAG RT was used to transfer the model to the Simulink environment. The concept of developing motors in JMAG 2D and 3D models and exporting in Simulink was developed and proven to be efficient⁵. In MATLAB/Simulink control system with cascaded closed torque and speed control loop is modeled for effective speed control for the imported model, one classic method is explained⁶. A new technology involving a one-loop PID controller utilizing neural network technology to enhance precision in determining the exact position of maximum efficiency is implemented⁷. As discussed previously, soft computing techniques offer good accuracy for PWM technique control, such as grey wolf optimization of the controller⁸, and can be enhanced with the Internet of Things (IoT) for remote access to the BLDC motor, as discussed⁹. Finally, it can be concluded that JMAG provides a platform for efficient motor design. Additionally, interfacing with JMAG and the RT lab to create a Hardware-in-the-loop environment is also possible, offering an innovative solution to the problem¹⁰.

The BLDC motor is initially designed in the JMAG environment using Finite Element Analysis (FEA), where electrical, magnetic, and thermal parameters are defined based on design specifications. Electromagnetic transient and coupled thermal analyses are carried out to obtain accurate motor characteristics. The validated FEA model is then exported using JMAG-RT as an S-function, enabling seamless integration with the MATLAB/Simulink platform. Further, the interaction between the exported real-time model and the cascaded control architecture is elaborated, including signal flow, parameter interfacing, and execution within the simulation environment. These additions provide a clear understanding of the modeling workflow and highlight the extent of the work carried out in this study.

In this study, a BLDC Motor model designed in JMAG to meet the required specifications and undergo electromagnetic transient analysis, coupled with iron loss analysis, was conducted to verify the proposed design terminology¹¹. The optimal design from JMAG was exported to MATLAB using JMAG-RT¹². For the exported model, the BLDC motor speed was regulated using a cascaded speed and voltage control loop designed with a PID controller¹³. This method is advantageous

because it addresses electromagnetic constraints; therefore, more accurate results are obtained compared to the conventional method, which uses the fundamental Eqn. for BLDC motor design. The system stability and response are governed by the duty cycle of PWM pulses generated with a control loop feedback¹⁴. The studies were conducted at variable load torques, and the corresponding speed was monitored using Matlab/Simulink¹⁵.

2. MODELLING OF BLDC MOTOR

An 8 pole and 12 slot Interior Permanent Magnet BLDC motor was designed in JMAG for specification, as shown in Table 1.

Table 1. BLDC motor designed specification

Design parts	Parameters	Values
Electrical parameters	Motor input power	152.02 Watts
	Rated current	5.429 Amps
	Resistance of each phase	1.031 ohms
	E.M.F. constant	0.2785 V/rad
Rotor specification parameters	Length of the motor	0.09 meter
	Dia. of rotor	0.0263 meter
	Inner pole diameter	20.53 mm
	No. of poles	8 Poles
	MMF required for flux	318.301 AT
Stator specification parameters	Flux	0.000743 wb
	Length of the magnet	2.865 m
	Inner diameter of stator	0.0273 m
	No. of conductors	442
	No. of slots	12
Inner dimension parameters	Stator back height	2.676 mm
	Height of the slot	9.193 mm
	Width of the teeth	0.001784 m

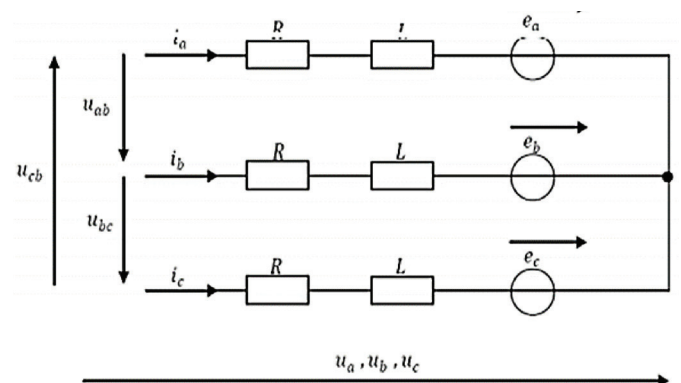


Figure 1. Equivalent electrical modeling of a BLDC motor.

Other parameters, such as motor inner and outer dimensions and electrical parameters, are calculated with Eqn. presented in¹⁶, and the calculated parameters in Table 1 are used in the design of the motor. The software provides a platform for iterative design for an optimal motor with reduced time consumption. Appropriate electrical and mechanical constraints were added to the design based on IEC standards, and Finite

Element Analysis (FEA) was carried out¹⁷. Electromagnetic transient analysis was performed once the motor model was established to determine its output characteristics and losses. Coupled thermal analysis was conducted, and once the results obtained were within the required tolerance, the model was finalized in JMAG. The fundamental Eqn. for the BLDC motor drawn from the equivalent circuit is as follows:

The following Eqn. describe the equivalent circuit,

$$v_a = R(i_a) + L \frac{d}{dt}(i_a) + e_a \quad (1)$$

$$v_b = R(i_b) + L \frac{d}{dt}(i_b) + e_b \quad (2)$$

$$v_c = R(i_c) + L \frac{d}{dt}(i_c) + e_c \quad (3)$$

Thus,

$$v_a = R(i_a - i_b) + L \frac{d}{dt}(i_a - i_b) + e_a - e_b \quad (4)$$

$$v_b = R(i_b - i_c) + L \frac{d}{dt}(i_b - i_c) + e_b - e_c \quad (5)$$

$$v_c = R(i_c - i_a) + L \frac{d}{dt}(i_c - i_a) + e_c - e_a \quad (6)$$

Where,

R = Armature Resistance

M = Mutual Inductance, which describes the flux linkage between two winding

L_s = Armature Inductance

$e_{a,b,c}$ = armature currents flowing through windings

$V_{a,b,c}$ = phase voltages

$V_{a,b}$ = phase-to-phase voltage

The relationship between phase currents,

$$i_a + i_b + i_c = 0 \quad (7)$$

Because each voltage is a linear combination of the other two, only two Eqn. are necessary. One variable was eliminated by omitting one Eqn..

$$v_{ab} = R(i_a - i_b) + L \frac{d}{dt}(i_a - i_b) + e_a - e_b \quad (8)$$

$$v_{bc} = R(i_a + 2i_b) + L \frac{d}{dt}(i_a + 2i_b) + e_b - e_c \quad (9)$$

According to Newton's second law of motion, the relationship between the electromagnetic torque of the motor and its speed can be expressed as follows:

$$T_E(t) - T_L(t) = J \frac{d(t)}{dt} + B^*(t) \quad (10)$$

$$i_a + i_b + i_c = 0 \quad (11)$$

$$\omega_m = \frac{d\theta_m}{dt} \quad (12)$$

where,

T_L = is the Load torque Nm

J = is the Rotor inertia in kgm^2

B = is the Damping constant

The equivalent electrical model of the BLDC motor has been revised to improve clarity, with enhanced labeling and clear representation of phase voltages, currents, and back-EMF components. Additionally, the dynamic behavior of the motor has been elaborated by expanding Eqn. (8), (9), and (10) to explicitly represent the time domain relationships between voltage, current, back EMF, and electromagnetic torque. These Eqn. are now presented in a more detailed form to capture the motor dynamics more accurately, thereby strengthening the theoretical foundation of the model and improving the overall understanding of the system operation. Eqn. (12) it is now explicitly identified as the angular velocity (Eqn. of the BLDC motor, representing the relationship between electromagnetic torque, load torque, and mechanical dynamics¹⁸.

2.1 JMAG RT

JMAG-RT is a tool used to link the JMAG motor model with MATLAB. Using this tool, the motor model designed with the required electrical and mechanical aspects is exported from JMAG to the MATLAB environment. The model is exported as an S function and is easily executable with reliable accuracy, since the FEA model provides more insight into motor characteristics than an equivalent mathematical model of the motor. FEA motor model created for electromagnetic transient analysis is as delineated in Fig. 2. Figure 2 represents the FEA-based BLDC motor model developed in the JMAG environment, highlighting key components such as the stator, rotor, permanent magnets, windings, and air gap.

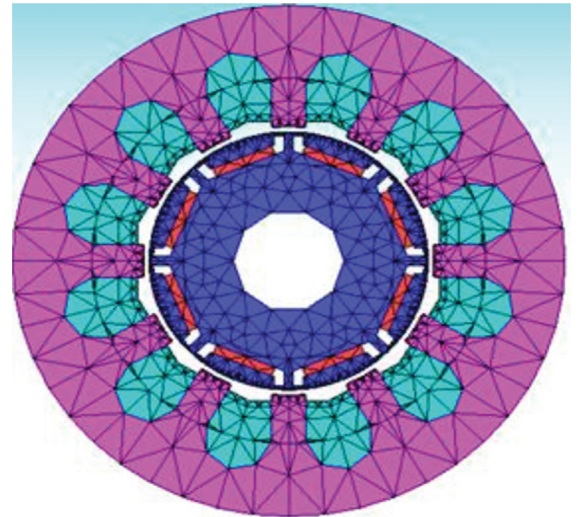


Figure 2. BLDC motor modeled using FEA.

2.2 Regulation of BLDC motor operation

The BLDC-RT motor model was driven using a voltage source inverter operating in 120 degree conduction mode. A schematic of the proposed system is shown in Fig. 3.

Here, a two-loop feedback controller is proposed with a novel PID controller for a simple and efficient control strategy: one loop is used as the speed control, and the other is for voltage control of the buck converter, which is cascaded as the inner feedback loop operates at a higher speed than the outer loop. A Hall sensor is employed to determine the motor's position for gate pulse generation in the PWM inverter. Two signal pulses are generated, one set is used to drive the DC-DC converter

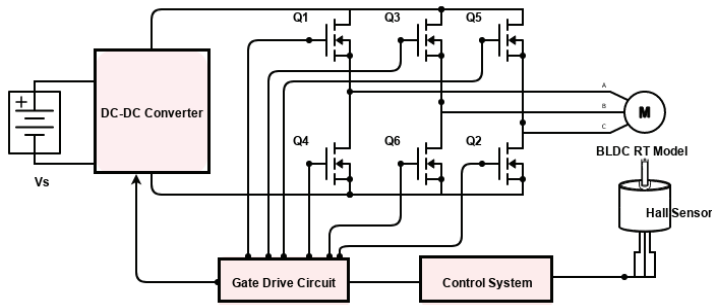


Figure 3. Schematic representation of the BLDC motor drive.

circuit, and another pulse is used for the inverter circuit. It now explicitly describes the cascaded dual loop control strategy, where the outer loop manages speed control and the inner loop regulates voltage via a DC–DC converter. The explanation includes how feedback from Hall sensors is used for rotor position detection, generation of PWM gate signals, and how the control loops interact to maintain stable speed under varying load conditions. This provides a clear understanding of the operational regulation and the effectiveness of the proposed control approach.

3. PID CONTROLLER

PID control technique is used here due to its accuracy among traditional controller. In this proposed technique, this well-known technique used due to its simplicity compared to complex control techniques used nowadays. It has a fast response to dynamics, with minimal and stable oscillation. Owing to its derivative gain in addition to the PI controller, the overshoot during the control id is reduced with oscillations from the output of the system. Therefore, the proposed system

of two cascaded controller loops uses two PID controllers, and the tuning method used to tune is a trial and error method.

4. SIMULATION MODEL FOR SPEED CONTROL

The simulation for the BLDC motor speed control is developed in the MATLAB environment, as illustrated in Fig. 4, BLDC motor RT, which is exported from JMAG-RT, and the speed control technique for the cascaded feedback loop is also modeled for export to control the motor RT model. The simulation model of the BLDC motor speed control is now clearly labeled with input signals, control loops, and motor model connections.

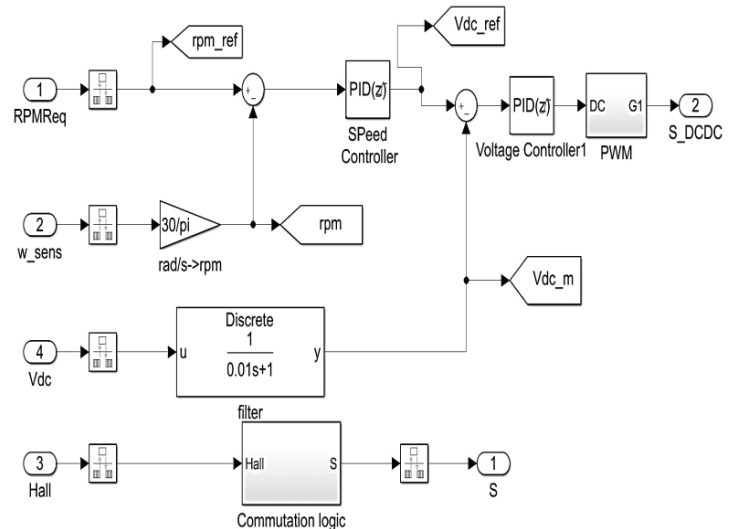


Figure 5. Cascade control of speed control.

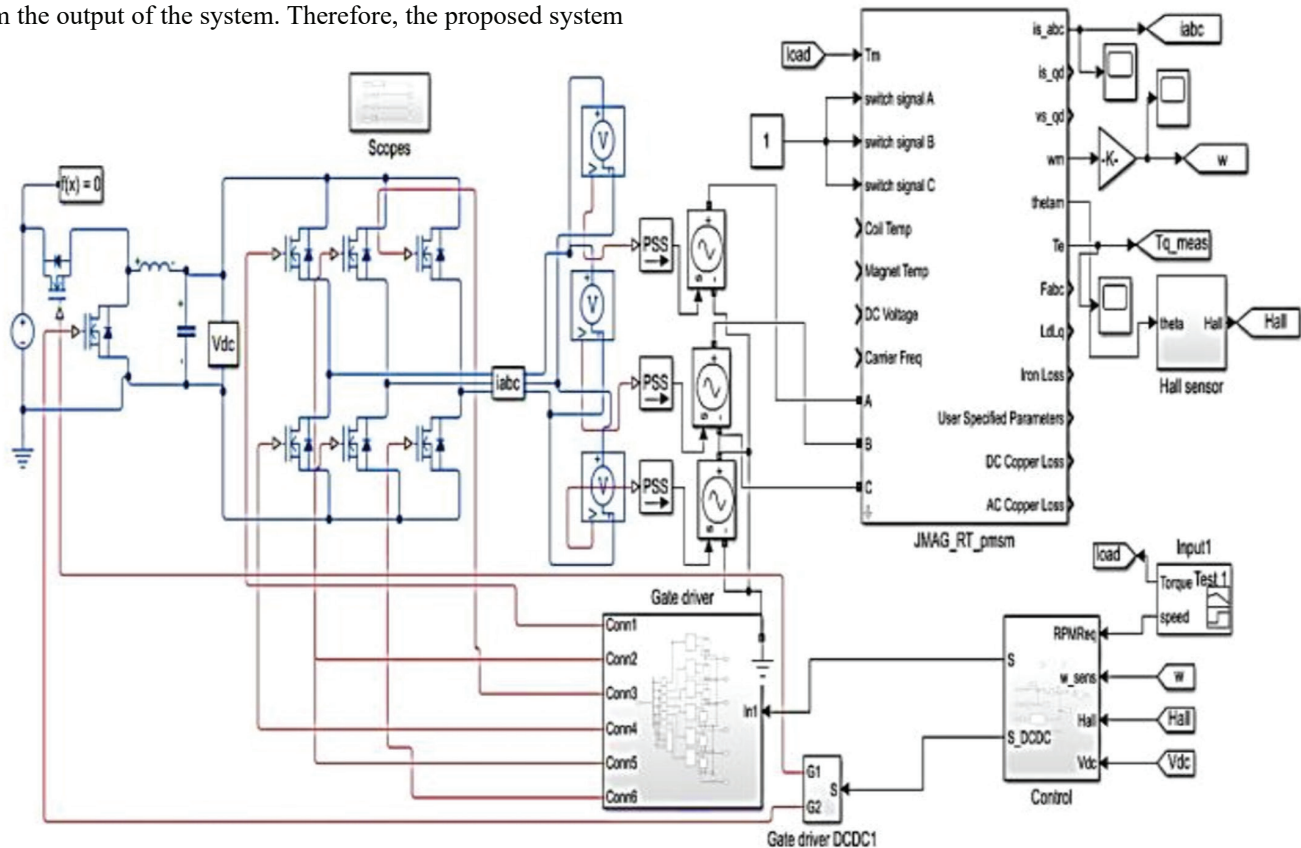


Figure 4. Closed loop speed control simulation model of the BLDC motor.

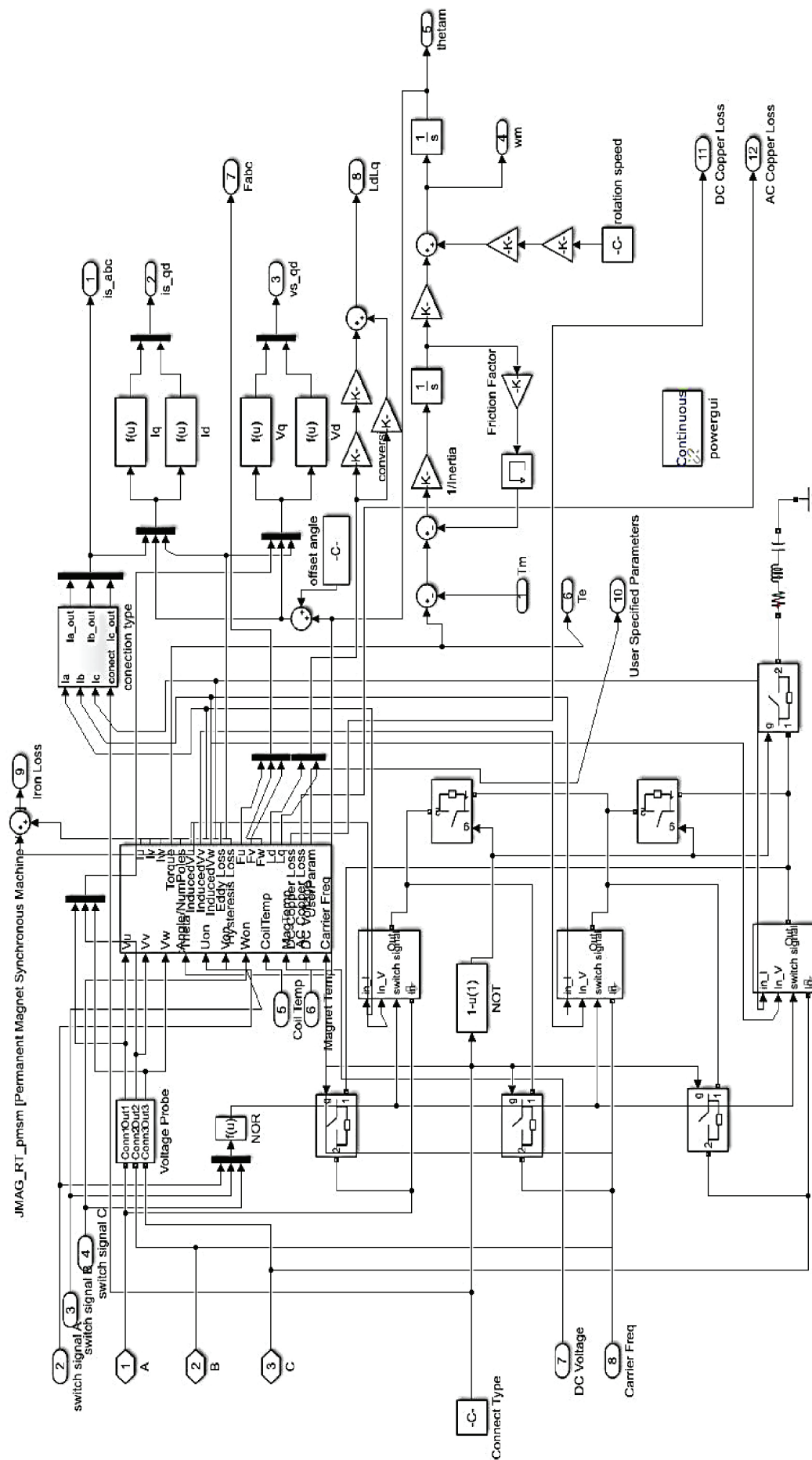


Figure 6. JMAG-RT simulink model.

The cascaded control loop is modeled as shown in the Fig. 4, with the outer loop managing speed and the inner loop controlling the voltage. The inner loop dynamics were faster than those of the outer loop. Both PID controllers were tuned using a trial-and-error method, and appropriate constant values were recorded for stable tuned output speed control. Figure 5 shows the connection between the speed controller (outer loop) and the voltage controller (inner loop) has been verified to show the cascaded reference signal flow.

The inverter used was the VSI, and the BLDC motor from the exported JMAG model was developed as the S function in the control process. The technique is efficient but complex, and is designed to export the designed motor model. The JMAG model exported as an RT model is shown in Fig. 6. It illustrates the high fidelity JMAG-RT motor model as integrated into the MATLAB/Simulink environment. Unlike a conventional mathematical model based purely on fundamental Eqn., this S function block incorporates the comprehensive electromagnetic and thermal characteristics derived from the Finite Element Analysis (FEA) performed in JMAG.

To address the coupling between electromagnetic and thermal domains, the S-function model exported from JMAG-RT represents a high-fidelity physics-based formulation incorporating both effects. The electromagnetic behavior of the BLDC motor is governed by the stator voltage Eqn. (13).

$$V = Ri + \frac{d\lambda}{dt} \quad (13)$$

where, the flux linkage λ is a nonlinear function of current and rotor position obtained from FEA. The electromagnetic torque is expressed as:

$$Te = \frac{P}{2}(\lambda i) \quad (14)$$

The thermal model is coupled through the loss components, where total losses are given by:

$$P_{loss} = I^2 R(T) + P_{core} + P_{mech} \quad (15)$$

Here, the stator resistance is temperature dependent and expressed as:

$$R(T) = R_0[1 + \alpha(T - T_0)] \quad (16)$$

The thermal dynamics are governed by:

$$C_{th} \frac{dT}{dt} = P_{loss} - \frac{(T - T_{amb})}{R_{th}} \quad (17)$$

These Eqn. establish the bidirectional coupling, where temperature affects electrical parameters (resistance, flux weakening), and electromagnetic losses contribute to heat generation. This coupled model is inherently embedded in the JMAG-RT exported S function block, as shown in Fig. 6, where the internal solver integrates electromagnetic field Eqn. with thermal network modeling.

In the Simulink environment, the S function block encapsulates these interactions, providing outputs such as torque, speed, current, and temperature dependent characteristics.

This ensures that the simulation reflects real time variations due to thermal effects, thereby improving the accuracy compared to conventional uncoupled mathematical models.

The DC converter used to reduce the complexity of the system, as mentioned earlier, has dynamics sufficient for voltage control for speed control feedback, and enhances the system's stability. The 120° mode of conduction is a power-efficient technique, as indicated by the graph in the simulation results. Table 2 represents the position sensing logic required to synchronize the stator's magnetic field with the rotor's position.

Table 2. Hall decoder truth table

Hall sensor-1	Hall sensor-2	Hall sensor-3	Back EMF-1	Back EMF-2	Back EMF-3
1	0	0	+1	-1	0
1	1	0	+1	0	-1
0	1	0	0	+1	-1
0	1	1	-1	+1	0
0	0	1	-1	0	+1
1	0	1	0	-1	+1

The PWM inverter uses the truth table shown in Table 3 to provide the gate pulse for all six MOSFET switches for Q1 to Q6, which is efficient when the conduction angle is chosen from 10° to 80°. Table 3 represents the execution logic for the Voltage Source Inverter (VSI) to drive the motor.

Table 3. Gate signal truth table

Ea	E _b	E _c	Q1	Q2	Q3	Q4	Q5	Q6
+1	-1	0	1	0	0	1	0	0
+1	0	-1	1	0	0	0	0	1
0	+1	-1	0	0	1	0	0	1
-1	+1	0	0	1	1	0	0	0
-1	0	+1	0	1	0	0	1	0
0	-1	+1	0	0	0	1	1	0

4.1 Simulation Under Opposing Loads

The speed output of the actual and reference speeds given to the BLDC motor is shown in Fig. 7 for a load torque of TL 1 Nm, and a rated speed of 670 RPM is generated with the model. The motor circuit current was approximately 3A for each phase, which shows the minimum current generated for the required speed and torque output. Figure 7 illustrates the tracking performance of the BLDC motor by comparing

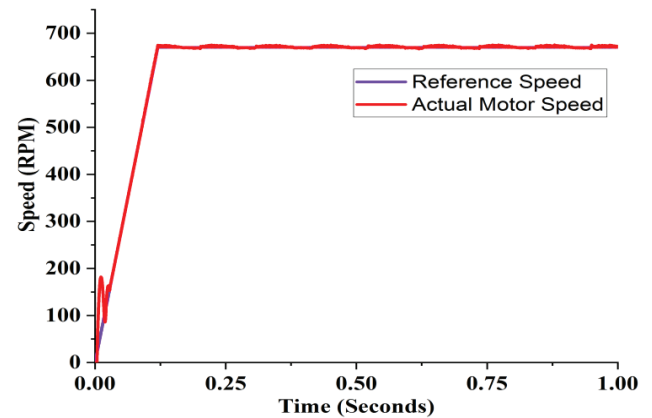


Figure 7. Rotor speed at load condition.

the Reference Speed and the Actual Motor Speed. This graph is critical for validating the effectiveness of the cascaded PID control architecture.

As shown Fig. 7, the red curve is the actual motor response speed output, which is the reference speed input, and is matched with the blue curve. The result thus obtained has a minimal ripple and negligible peak overshoot and rise time. The motor is designed for a load of 1 Nm torque, which is satisfied by the proposed model, as shown in Fig. 8. The graph shows that the motor successfully generates the required torque to meet the load demand of 1 Nm.

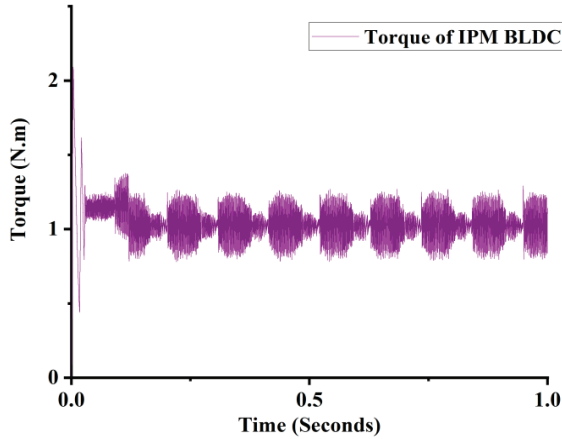


Figure 8. Output waveform of the torque of the motor.

The three-phase output current waveform, as shown in Fig. 9, has a maximum of approximately 3 A, and the waveform is a disturbed trapezoidal waveform. The current thus obtained is well below the rated current, so the thermal stability of the motor is maintained. Maintaining the current at this level ensures thermal equilibrium. By keeping the current below the rated threshold, the simulation proves that the motor can operate continuously without overheating the windings or demagnetizing the permanent magnets, which is a critical requirement for aerospace and robotic applications.

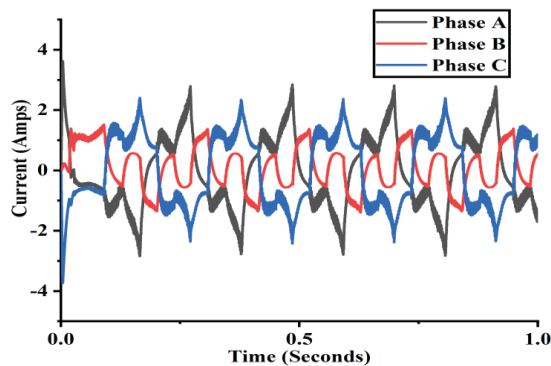


Figure 9. Output waveforms of the currents.

4.2 Performance Analysis under Varying Speed and Step Commands

The dynamic performance of the proposed FEA-based BLDC motor model was evaluated through a series of simulated tests involving step changes in speed and varying command inputs. This analysis ensures the cascaded PID

control structure maintains stability and precision across the motor's operational envelope.

4.2.1 Step Speed Command Analysis

When a step increase in the reference speed is applied, the system demonstrates the following characteristics:

- **Rise Time and Settling Time:** The motor reaches the rated speed of 670 RPM within approximately 0.1 seconds, showing a fast response to dynamics
- **Overshoot Management:** Due to the derivative gain in the PID controllers, the peak overshoot is negligible, preventing mechanical stress during sudden acceleration
- **Steady State Error:** The actual motor speed (red curve) aligns precisely with the reference blue curve, indicating near zero steady-state error under closed-loop regulation.

4.2.2 Response to Varying Load Torques

The robustness of the velocity regulation was tested by applying a load torque of 1 Nm.

- **Torque Ripple:** The output waveform of the torque exhibits minimal ripple, which is critical for applications like electromechanical actuators (EMA) in aircraft or robotics
- **Current Regulation:** Under these varying conditions, the three-phase current remains at approximately 3 A, which is well below the rated current of 5.429 A
- **Thermal Stability:** Maintaining the current below the rated limit ensures that thermal equilibrium is achieved, guaranteeing the motor's lifespan.

4.2.3 System-Level Validation

- **Control Synergy:** The inner voltage control loop operates at a higher speed than the outer speed loop, allowing the buck converter to respond rapidly to fluctuations caused by the 120° conduction mode
- **Efficiency:** The use of JMAG-RT allows the simulation to capture electrical and mechanical constraints that first-principles Eqn. omit, leading to results that are more reliable for hardware-in-the-loop (HIL) testing. Table 4 shows summary of dynamic performance metrics.

Table 4. Dynamic performance metrics

Parameter	Observed Value	Significance
Rated speed	670 RPM	Matches design specification
Settling time	~0.1 sec	High-speed response for EMAs
Peak overshoot	Negligible	Stable PID tuning
Operating current	~3 A	Maintains thermal equilibrium
Load torque	1 Nm	Validates high-fidelity FEA model

4.3 Additional Test Case Validation

To further evaluate the effectiveness and robustness of the proposed FEA based BLDC motor model, additional test cases were carried out under different operating conditions.

Case 1: Variable Load Torque Condition

The load torque applied to the motor was varied from 0.5 Nm to 1.5 Nm. The simulation results indicate that the proposed cascaded control structure maintains stable speed regulation with minimal deviation. The motor quickly compensates for load disturbances, demonstrating strong disturbance rejection capability.

Case 2: Step Change in Reference Speed

A step variation in reference speed from 400 RPM to 670 RPM was applied. The system exhibits a fast transient response with negligible overshoot and reduced settling time. The smooth transition confirms the effectiveness of the PID-based cascaded control in handling rapid command changes.

Case 3: Sudden Load Disturbance

A sudden load disturbance of 1 Nm was introduced during steady-state operation. The motor speed initially shows a slight dip but rapidly recovers to the reference value, confirming the robustness of the control strategy.

Case 4: Low-Speed Operation

The motor was tested under low-speed conditions (below 300 RPM) to evaluate controllability. The results show stable operation without significant oscillations, validating the suitability of the proposed model for a wide operating range.

These additional test cases clearly demonstrate that the proposed system provides reliable performance under dynamic and uncertain operating conditions. The inclusion of electromagnetic and thermal effects through the JMAG-RT model further enhances the accuracy and practical applicability of the simulation results.

5. CONCLUSION

Modeling of the BLDC motor proposed with electrical and mechanical constraints is carried out for a loading condition of 1 Nm torque, which is modeled in JMAG and exported. Speed control with a PID control cascaded two-loop feedback controller with a DC-DC converter and VSI was modeled in a MATLAB environment. The proposed control technique is used to control the exported motor model, and because the motor is FEA analyzed with thermal and mechanical aspects, the proposed model is more reliable than the mathematical model.

Thus, the results obtained have been verified with the generated output graphs, and the motor with control is thus created and used for applications sensitive to electrical and mechanical aspects, such as thermal withstanding capability and loss distribution. Because of the 120° mode of control with the DC-DC converter for speed control, the power efficient model achieved as the current through is 3 A, which is below the calculated rated current of 5 A, which indicates that thermal equilibrium is achieved; thus, the life span of the motor is guaranteed. The future scope of the proposed concept is that the model can be used for controlling with soft computing techniques and can be verified with the efficiency obtained with the proposed complexity. In summary, the integration of FEA-based motor parameters into a dynamic

control simulation provides a superior platform for Hardware-in-the-Loop (HIL) testing. These simulated results confirm that the designed BLDC motor meets the stringent dynamic and thermal requirements of electromechanical actuators, offering a highly accurate precursor to physical prototyping.

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